

Hypothesis Testing

Errors, Power, Likelihood Ratios, and Multiple Comparisons

Hypothesis Testing Framework

Definition — Statistical Hypothesis Test

A **statistical hypothesis test** is a procedure for deciding between two competing claims about a population parameter θ .

Null Hypothesis

$$H_0 : \theta \in \Theta_0$$

The hypothesis to be tested — the status quo assumption.

Alternative Hypothesis

$$H_1 : \theta \in \Theta_1$$

The research hypothesis — what we want to detect.

Mathematical Requirements

- **Mutually exclusive:** $\Theta_0 \cap \Theta_1 = \emptyset$
- **Exhaustive:** $\Theta_0 \cup \Theta_1 = \Theta$ (the complete parameter space)

Types of Hypotheses

Simple hypothesis — specifies a single parameter value

$$H_0 : \theta = \theta_0$$

Composite hypothesis — specifies a range of values

$$H_0 : \theta \geq \theta_0$$

i Two-sided

$$H_0 : \theta = \theta_0$$

$$H_1 : \theta \neq \theta_0$$

Deviations in
either direction.

i Right-tailed

$$H_0 : \theta \leq \theta_0$$

$$H_1 : \theta > \theta_0$$

Increases
above θ_0 .

i Left-tailed

$$H_0 : \theta \geq \theta_0$$

$$H_1 : \theta < \theta_0$$

Decreases
below θ_0 .

Rejection Regions, Live

The alternative decides **where** the rejection region goes; α decides **how big** it is.

Alternative

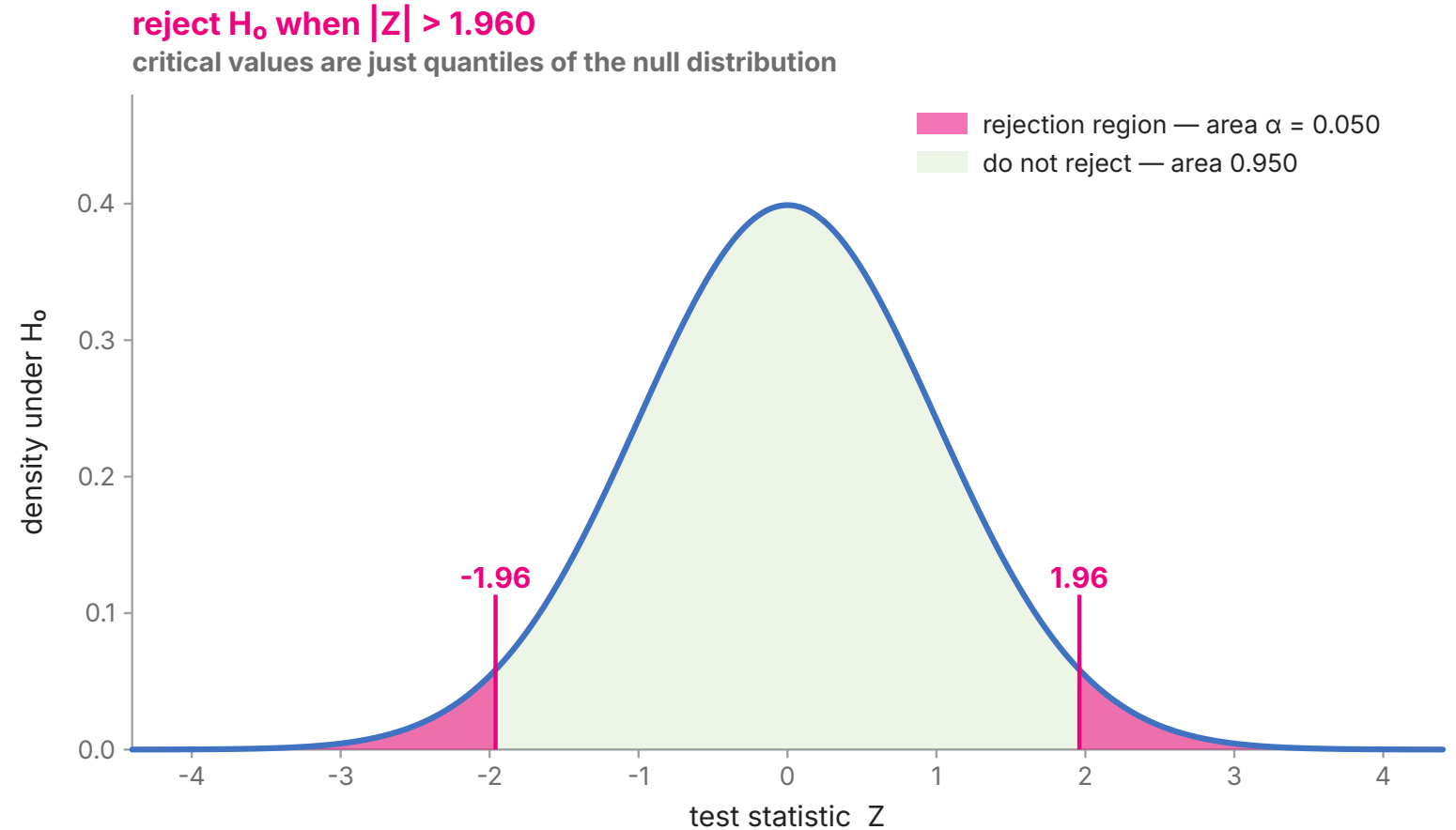
Two-sided $H_1: \theta \neq \theta_0$ ▼

Significance α

0.05



The curve is the distribution of the test statistic **under H_0** . The shaded area *is* α — the probability of rejecting a true null.



Test Statistics and Decision Framework

Definition — Test Statistic

A **test statistic** $T(X)$ is a function of the sample data used to decide between H_0 and H_1 .

Decision Rule

A test is defined by its **critical region** (rejection region) C :

Reject H_0 if $T(X) \in C$

Critical Value

The boundary value c such that

$$P(T(X) > c \mid H_0) = \alpha$$



Significance Level

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true})$$

Common choices: 0.01, 0.05, 0.10.

Errors and Power

Type I and Type II Errors

Decision	H_0 true	H_0 false
Reject H_0	Type I error — probability α	Correct decision — power $1 - \beta$
Fail to reject H_0	Correct decision — probability $1 - \alpha$	Type II error — probability β

Error Definitions

$$\alpha = P(\text{Reject } H_0 \mid H_0 \text{ true})$$

$$\beta = P(\text{Fail to reject } H_0 \mid H_1 \text{ true})$$

The Trade-off

- Decreasing α **increases** β — at fixed sample size
- Only a larger n reduces both at once

Two Errors, One Threshold, Live

One threshold splits both curves at once: you cannot shrink one shaded area without growing the other.

Alternative

Right-tailed ▾

Effect size $\delta = (\mu_1 - \mu_0)/\sigma$

0.5

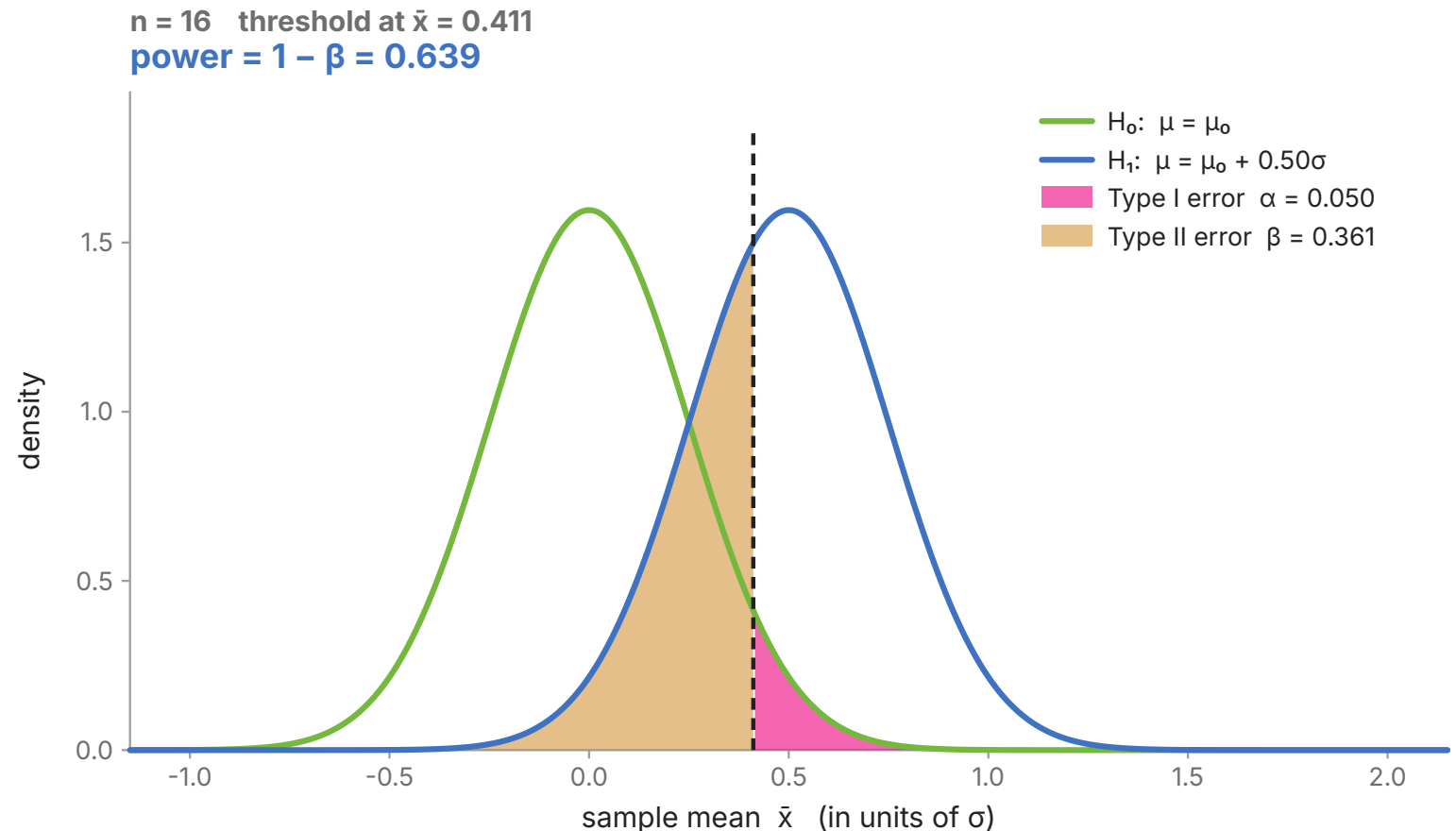
Sample size n

16

Significance α

0.05

Both curves are the sampling distribution of $\bar{X}$, under H_0 and under H_1 . Raising n narrows **both** — the only move that shrinks α and β together.



Statistical Power

Definition — Power

The **power** of a test is the probability of correctly rejecting H_0 when it is false:

$$\text{Power} = 1 - \beta = P(\text{Reject } H_0 \mid H_1 \text{ true})$$

Power Function

$$\pi(\theta) = P_{\theta}(\text{Reject } H_0)$$

How power varies with the **true** parameter value.



Properties

- $\pi(\theta) \leq \alpha$ for $\theta \in \Theta_0$
- π grows as θ moves away from θ_0 **towards** H_1
- π grows with n



Factors

- Sample size $n \uparrow$
- Significance level $\alpha \uparrow$
- Effect size $\uparrow$
- Population variance $\downarrow$

The Power Curve, Live

Each curve is one sample size. Where they cross θ_0 , every curve is pinned at α .

Alternative

Two-sided ▾

Significance α

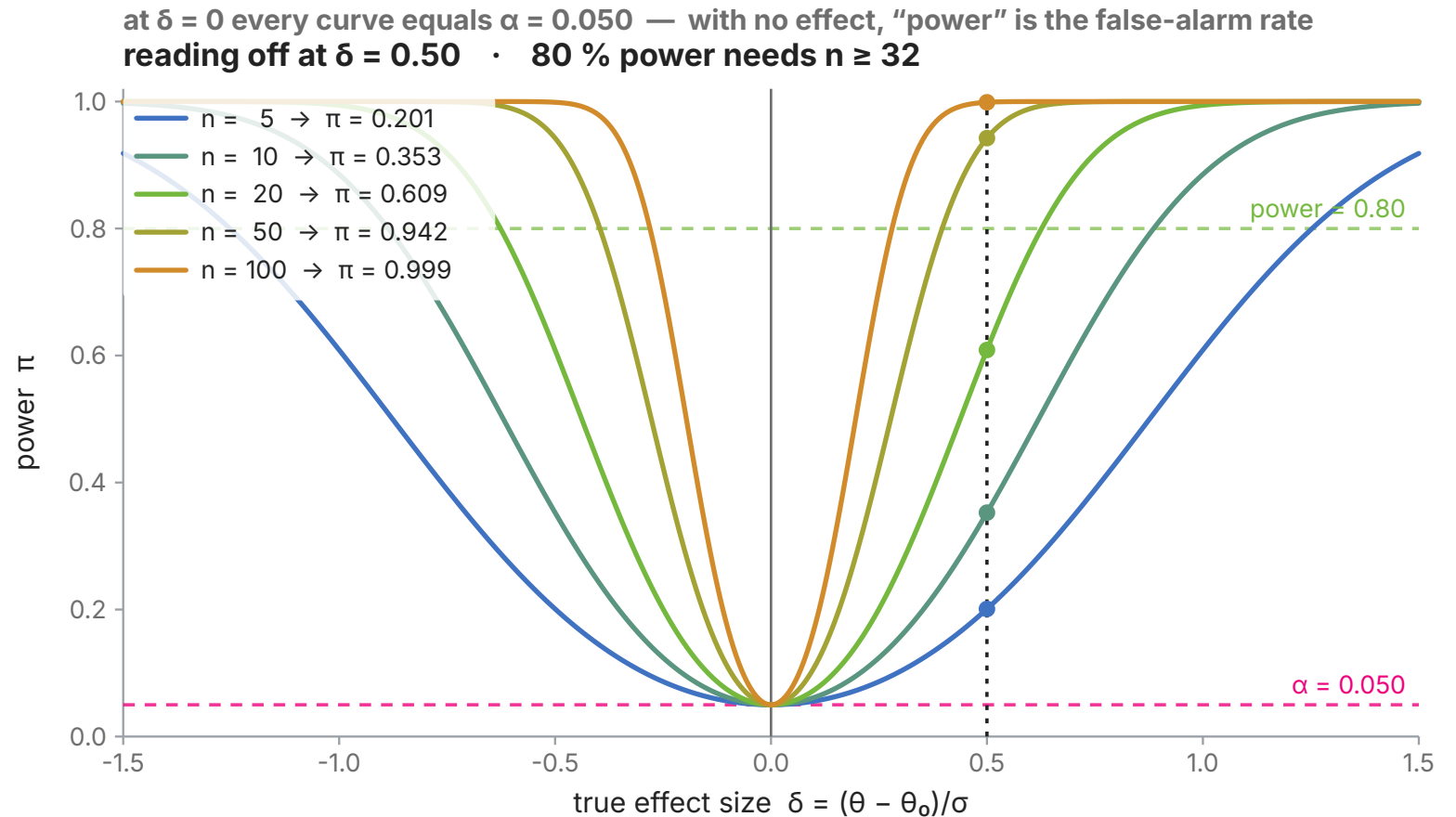
0.05

Read off at $\delta =$

0.5

Plot against $\sqrt{n} \cdot \delta$ ☐

$\delta = (\theta - \theta_0)/\sigma$ is the true effect in standard deviations; $\pi(\delta)$ is the blue area beyond the threshold on the *Two Errors* slide. Known- σ z -test: with σ estimated (t -test), power at small n is somewhat lower.



Reading the Power Curve

- **One formula** (right-tailed): $\pi(\delta) = 1 - \Phi(z_{1-\alpha} - \sqrt{n} \delta)$. The effect only enters through the *signal-to-noise ratio* $\lambda = \sqrt{n} \delta$ — how many standard errors the truth sits away from θ_0 .
- **The $\sqrt{n}$ law.** To detect an effect half as large, you need **four times** the data. Precision is expensive.
- **Planning a study.** Solving $\pi = 1 - \beta$ for n : $n \approx \left(\frac{z_{1-\alpha/2} + z_{1-\beta}}{\delta} \right)^2$. For $\delta = 0.5$, $\alpha = 0.05$, 80 % power: $n \approx ((1.96 + 0.84)/0.5)^2 \approx 32$.
- **One-sided buys power in one direction** by giving up all power in the other: the curve is higher on its side of θ_0 and falls *below* α on the other side.

! The point

"Not significant" from an underpowered study is **not** evidence of no effect. At $n = 10$ and $\delta = 0.5$, the test misses a real effect about two times out of three.

Likelihood Ratio Tests

Likelihood Ratio Test (LRT)



Definition — Likelihood Ratio Statistic

$$\Lambda(x) = \frac{\sup_{\theta \in \Theta_0} L(\theta; x)}{\sup_{\theta \in \Theta} L(\theta; x)}$$



LRT Decision Rule

Reject H_0 if $\Lambda(x) < c$, with c chosen so that

$$P(\Lambda(X) < c \mid H_0) = \alpha$$



Theorem — Wilks

Under regularity conditions, as $n \rightarrow \infty$,

$$-2 \log \Lambda(X) \xrightarrow{d} \chi_k^2$$

where $k = \dim(\Theta) - \dim(\Theta_0)$.



Why it matters

- One recipe for a huge family of tests
- Asymptotically optimal
- p-values straight from a χ^2 table

The Likelihood Ratio, Live

— $2 \log \Lambda$ is exactly **twice the drop** in log-likelihood from its peak down to the best point allowed by H_0 .

Model

Normal mean, $\sigma = 1$ $H_0: \mu = 0$ ▾

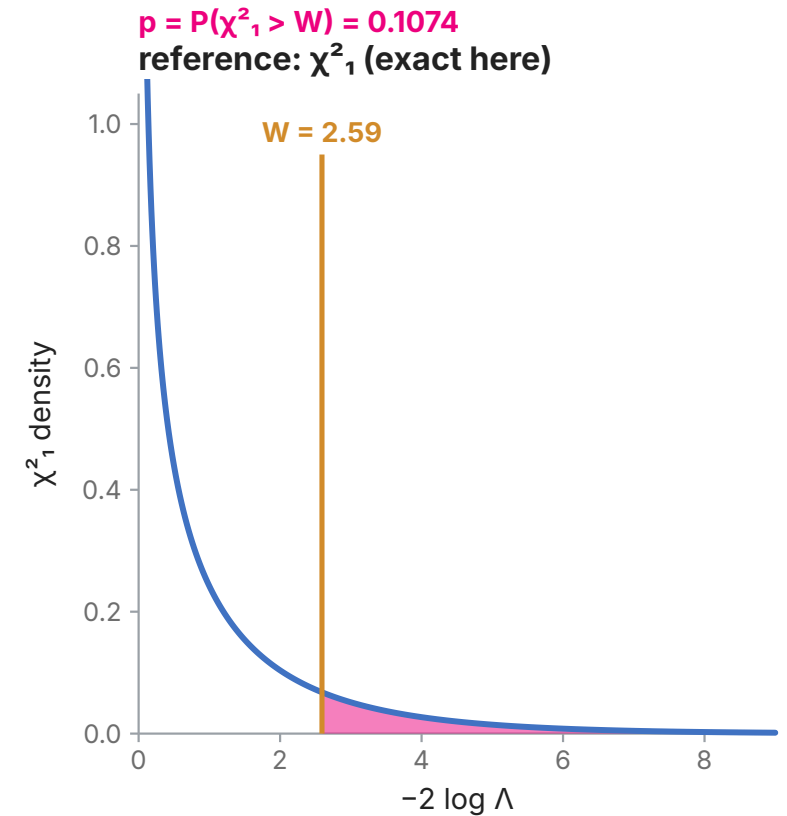
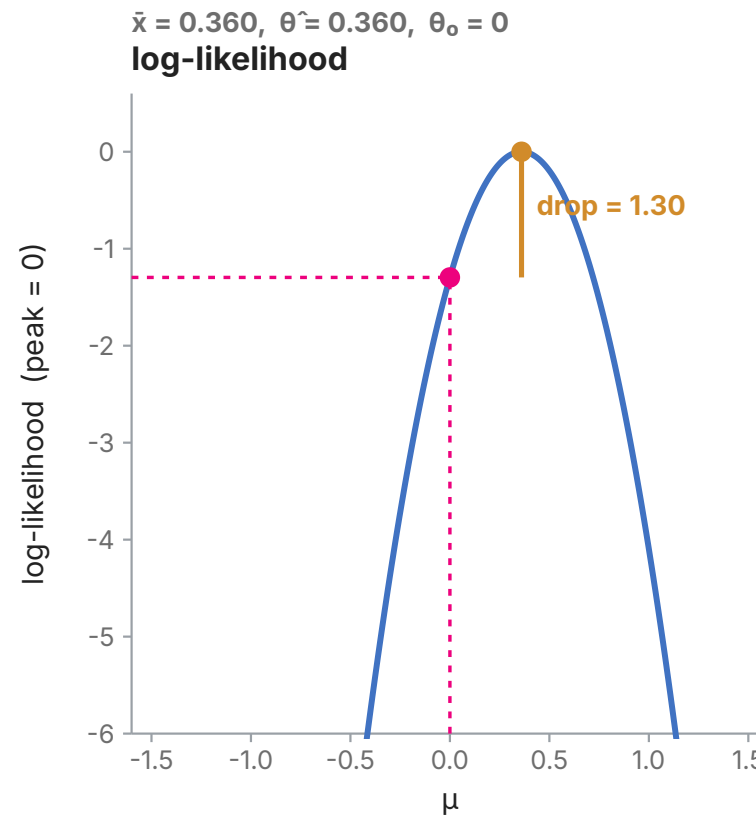
Sample size n

20

Observed statistic

0.62

Left: the log-likelihood, shifted so its peak is 0. Right: the χ^2_1 reference. For the normal mean the χ^2_1 law is **exact**; otherwise it is the large- n approximation.



P-values

P-values: Definition and Interpretation

Definition — p-value

The **p-value** is the probability of observing a test statistic as extreme or more extreme than the observed value, **assuming** H_0 is true:

$$p\text{-value} = P_{\theta_0}(T(X) \geq T(x_{\text{obs}}))$$

with T oriented so that **large = far from θ_0 towards H_1** . For $Z = \sqrt{n}(\bar{X} - \theta_0)/\sigma$: right-tailed $p = P_{\theta_0}(Z \geq z_{\text{obs}})$; two-sided $p = P_{\theta_0}(|Z| \geq |z_{\text{obs}}|) = 2(1 - \Phi(|z_{\text{obs}}|))$.



Correct Interpretation

- Small $p \Rightarrow$ evidence **against** H_0
- $p < \alpha \Rightarrow$ reject H_0 at level α
- p measures **compatibility** of the data with H_0



A p-value is NOT

- $P(H_0 \text{ true} \mid \text{data})$
- The probability of making an error
- A measure of effect size

A large p -value is **not** evidence that H_0 is true — only that the data do not contradict it.

What a p-value Actually Does, Live

Under H_0 the p-value is **exactly uniform**. Every property people expect of it follows from that one fact.

True effect δ

0



Sample size n

16

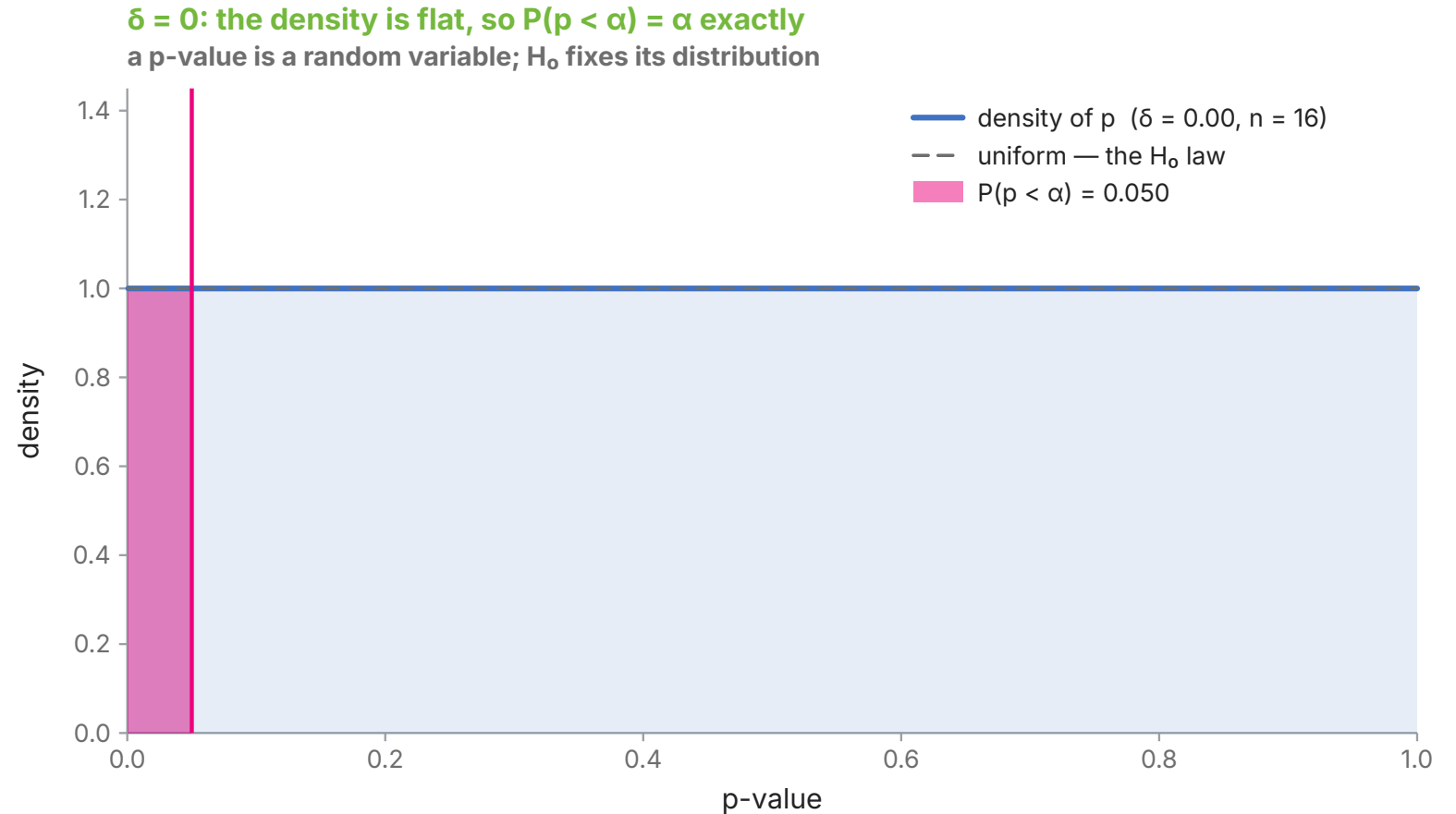


Significance α

0.05



Exact density of the two-sided p-value for a z -test. At $\delta = 0$ it is flat at 1 — so $P(p < \alpha) = \alpha$, which is what “level α ” means. Raising δ or n piles the mass up against 0.



Worked Examples

Example 1: Normal Mean Test



Problem Setup

Test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ for $X_1, \dots, X_n \sim N(\mu, \sigma^2)$.



Case 1 — Known Variance

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0, 1) \text{ under } H_0$$

Reject if $|Z| > z_{\alpha/2}$;
 $p = 2P(Z > |z_{\text{obs}}|)$



Case 2 — Unknown Variance

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \text{ under } H_0$$

with

$$S^2 = \frac{1}{n-1} \sum_i (X_i - \bar{X})^2.$$

Reject if $|T| > t_{n-1, \alpha/2}$;
 $p = 2P(t_{n-1} > |t_{\text{obs}}|)$

Why t , not z , Live

Estimating σ costs you something. The price is paid in the tails.

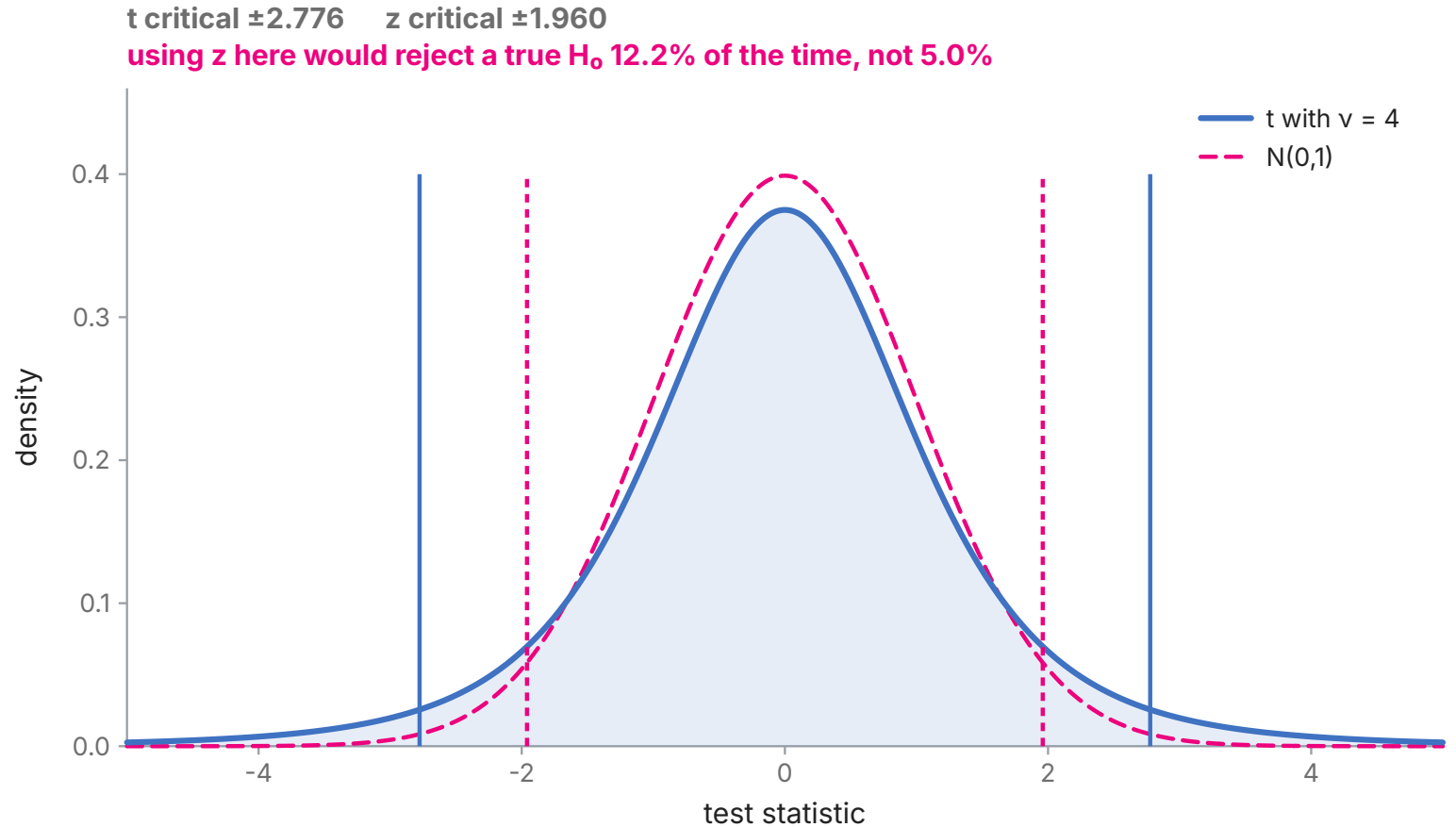
Sample size n

5

Significance α

0.05

Using $z_{\alpha/2}$ when σ is estimated does not give you level α — it gives you the larger error rate in the readout. The gap closes as $n \rightarrow \infty$.



Example 2: Variance Test



Problem Setup

Test $H_0 : \sigma^2 = \sigma_0^2$ against $H_1 : \sigma^2 \neq \sigma_0^2$ for $X_1, \dots, X_n \sim N(\mu, \sigma^2)$.



Chi-Square Test

$$\chi^2 = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi_{n-1}^2 \text{ under } H_0$$

Reject if $\chi^2 < \chi_{n-1, 1-\alpha/2}^2$ **or**
 $\chi^2 > \chi_{n-1, \alpha/2}^2$.



Health Warning

- **Very sensitive** to the normality assumption
- Non-normal data can give badly wrong conclusions
- Prefer robust alternatives when in doubt

Note the null distribution is **skewed**, so the two critical values are not symmetric about $n - 1$ — unlike the z and t tests.

Multiple Testing

The Multiple Testing Problem

! The Problem

Running m independent tests at level α , the chance of at least one Type I error is

$$P(\text{at least one Type I error}) = 1 - (1 - \alpha)^m$$

With $\alpha = 0.05$ and $m = 20$: $P \approx 0.64$.

i Family-wise Error Rate

$$\text{FWER} = P(\text{reject at least one true } H_0)$$

Bonferroni: test each at α/m ,
which guarantees
 $\text{FWER} \leq \alpha$.

i False Discovery Rate

$$\text{FDR} = E \left[\frac{\# \text{ false discoveries}}{\# \text{ total discoveries}} \right]$$

Less conservative than FWER
control — Benjamini–
Hochberg.

Multiple Testing, Live

Every square is one test. The nulls are genuinely true — every red square is a mistake the procedure made.

Number of tests m

100



True effects

10



Level α

0.05



Correction

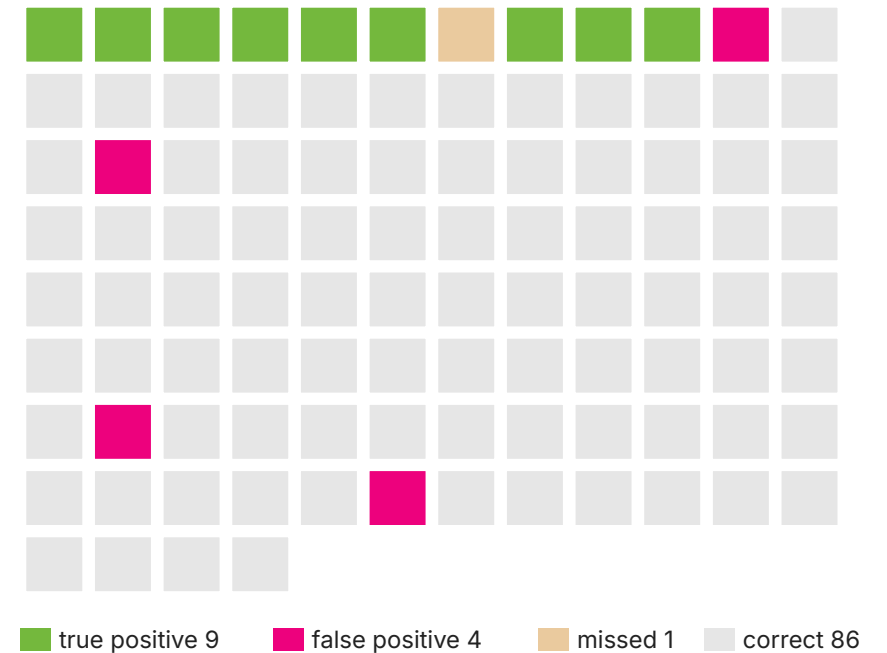
None



New draw



13 discoveries, 4 of them false — realised FDR 0.308
 $m = 100$ tests, 10 with a real effect



What the Grid Shows

- With **no correction** and 100 true nulls, you expect $100\alpha = 5$ false positives every single time — regardless of whether any effect exists.
- **Bonferroni** all but eliminates them, at the cost of missing real effects: watch the orange squares multiply.
- **Benjamini-Hochberg** sits in between — it lets a *controlled fraction* of discoveries be false, and keeps far more of the true ones.

! The point

There is no correction that is simply “best”. FWER and FDR answer different questions, and you must decide which error you can live with **before** looking at the data.

Summary

Key Takeaways

- (1) **Hypothesis testing** is a decision framework under uncertainty, not a proof procedure
- (2) **Type I and II errors** trade off against each other at fixed n — only more data improves both
- (3) **Power** is the probability of detecting a real effect, and it depends on n , α , effect size and variance
- (4) **P-values** measure evidence against H_0 ; under H_0 they are uniform, and that is all “level α ” means
- (5) **Multiple testing** inflates false positives unless you correct for it



The Test Toolkit

- **LRT** — general recipe, asymptotically optimal, $-2 \log \Lambda \rightarrow \chi_k^2$
- **Neyman-Pearson** — most powerful test for simple vs simple
- z -test — known variance
- t -test — unknown variance (the usual case)
- χ^2 test — variance testing



Critical Reminder

Statistical significance $\neq$ practical significance.



Best practice

Choose α and the sample size **before** seeing the data; report effect sizes alongside p-values; account for every test you ran, not just the ones that worked.